# Plant preprocessing benchmark evidence

Arabidopsis DAP-seq and bulk ATAC; matched-stage comparisons

October 16, 2026

**Abstract**

We reconcile 64 completed method/repetition cases across 6 public-data subsets. In a separate MACS3 test, all 24 CPU-specific comparisons preserved the evaluated peak outputs, with 1-CPU speedups of 4.20 to 6.32 times. Most package alternatives lack evidence of a consistent biological improvement, and the broader species panel remains incomplete. We would retain scientific defaults while evaluating the pinned optimized caller within its tested scope.

## Setup

**Data.** Two SE DAP treatments, four PE DAP treatments and two biological ATAC replicates were sampled at 1-million and 5-million template caps. DAP treatments are not extra biological replicates; repeated execution is not a new experiment.

**Scoring.** Peak Jaccard is intersecting divided by union peak bases. Held-out FRiP is the fraction of retained fragments overlapping the other biological replicate's peaks. Fixed-region Spearman measures count agreement, not truth. Independent PBM motif support uses matched windows; cost is secondary to these scientific comparisons.

**Table 1.** Completed method/repetition cases, not biological sample counts.

| Data | Treatment / replicate identity | Cases |
|---|---|---|
| SE DAP | FUS3, ABI5; assigned control | 16 |
| PE DAP | Four protocol-optimization treatments | 16 |
| Bulk ATAC | SRR4000468 and SRR4000469 | 32 |

## 1 Depth changes discovery more than these small package differences

For the 2 ATAC biological replicates, peak Jaccard increases from 0.2404 at 1 million pairs to 0.4858 at 5 million pairs. Held-out FRiP rises from 0.1433 to 0.4601 and from 0.1633 to 0.4023, while peak coverage also increases.

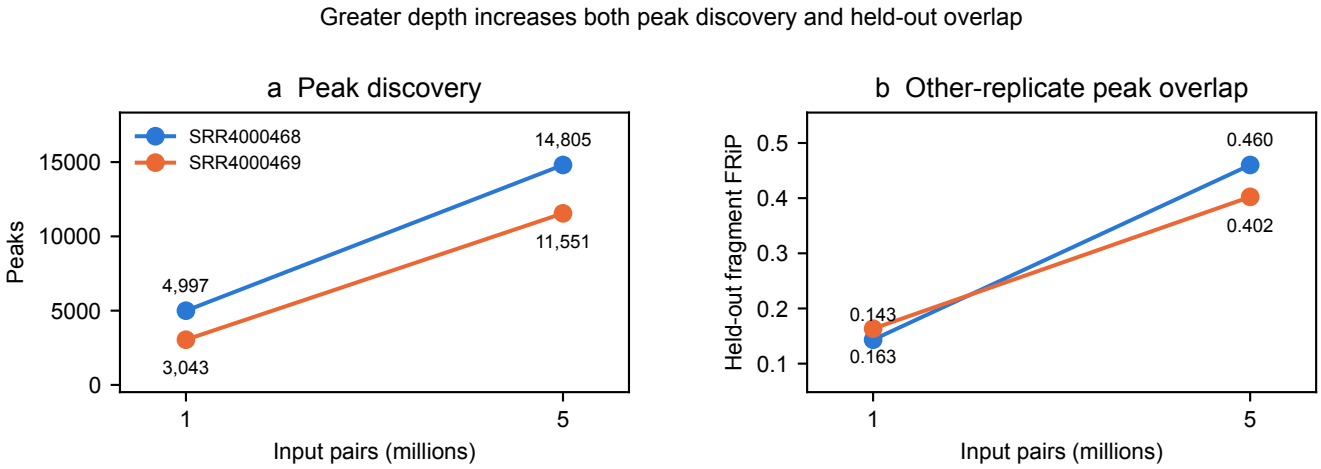



**Figure 1.** TAIR10 baseline peak discovery and cross-replicate overlap.

## 2 The current evidence supports scoped decisions

The 64 cases reconcile to the historical matrix count, but do not represent 64 independent biological experiments. A separate local audit found 8 completed frozen reference baselines and 3 completed assessments; these are not added to the method-matrix denominator.

**Table 2.** Decisions supported by the reports in this series.

| Step | Observed evidence | Decision |
|---|---|---|
| Alignment | BWA is close; Bowtie2 changes PE DAP peaks | Retain BWA-MEM2 |
| Trimming | Two short-read ATAC replicates change little | Defer general change |
| Duplicates | Picard is faster in one ATAC cohort | Broaden validation |
| Peaks | 24 exact-output optimized MACS3 comparisons | Scoped opt-in |
| Counting | Independent read-end arithmetic agrees | Retain implementation |
| References | High eligible-region concordance; exclusions remain | Choose explicitly |

**Table 3.** Coverage gaps are distinct from failed comparisons.

| Step / claim | Remaining evidence |
|---|---|
| QC and signal-generation packages | No completed matched-package ranking in this series |
| Other plant species | Acquisition or mapping checks do not establish method performance |
| Model training or sequence design | No benchmark here establishes predictive or experimental benefit |

## Open issues

- Subsets are not full-study validation; short libraries may reach full yield.
- Baseline agreement, own FRiP and peak counts are not ground truth.
- The nonexclusive host and CPU allocations limit timing transfer.

## Recommendation

We would use these reports to select the next acceptance tests, without changing scientific defaults.

1. Evaluate candidate changes on longer reads and additional plant genomes.
2. Require independent support and biological reproducibility before selecting a new default.

## Appendix: Methods

We summarize recorded results without rerunning biological processing. The main Arabidopsis matrix uses seeded nested raw-template subsets (seed 60141), three fresh technical repetitions at the 1-million cap and one at the 5-million cap. Actual counts replace requested caps for short libraries. DAP retains primary mapped reads including MAPQ 0; ATAC selects proper primary pairs with both mates MAPQ at least 30, excludes MAPQ 255, organelles and marked duplicates. Biological identity is preserved. Source files are SHA-256 checked against the accompanying catalog; evidence snapshot revision ed1c50b8e4b5 includes the October 13 to 16 records. Completed local reference assessments are separately identified in that catalog.